Convolutional Neural Networks for Fake News Classification

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Abstract

We apply convolutional neural networks to the classification of news articles as real or fake. A CNN over word embeddings reaches 93.1 percent accuracy on a public English dataset.

Keywords: fake news classification, convolutional neural network, word embeddings, text classification

1. Introduction

Fake news classification can be framed as a text classification problem. We ask whether convolutional filters over word embeddings capture the stylistic patterns that separate fake from real articles.

2. Related Work

Text CNNs have been used for sentiment analysis and topic classification. Few studies apply them to fake news.

3. Methodology

Articles are tokenised and mapped to 300-dimensional word embeddings. A CNN with three filter sizes extracts n-gram features, followed by max pooling and a dense classifier.

4. Results

The CNN reached 93.1 percent accuracy and an F1 score of 0.92. It trained in under one hour on a single GPU.

5. Limitations

The dataset is limited in size and contains only English articles. The model does not use social context.

6. Conclusion

CNNs are a simple and effective baseline for fake news classification.

7. Future Work

Future work should apply the model to multilingual datasets and larger corpora.

References

[1] J. Example, An Early Study of Linguistic Cues in Deceptive Text, Journal of Demonstration Studies, 2015.

Note: this is a SYNTHETIC demonstration paper with fictional authors and invented results, created to demo ResearchGraph. It is not a real publication.